

A Systematic Mapping Study of Human Centered Requirement Engineering for Artificial Intelligence

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Abstract—Human-Centered Artificial Intelligence (HCAI) emphasizes designing AI systems that augment human capabilities while upholding ethical, transparent, and value-sensitive principles. Requirements Engineering (RE) is well positioned to support this vision. To understand how RE currently contributes to HCAI, this paper presents a Systematic Mapping Study (SMS) of 31 peer-reviewed studies published since 2000. The review investigates four key dimensions: (1) which RE stages are addressed, (2) the human-centered processes, methods, techniques, and artifacts employed, (3) the non-functional requirements (NFRs) captured, and (4) the domains in which RE4HCAI is applied. Results reveal a strong emphasis on early-stage elicitation activities, with limited formalization, specification, and traceability of human-centered requirements. Usability, transparency, and explainability dominate the NFR landscape, while fairness, accountability, robustness, and other critical HCAI attributes remain inconsistently defined and infrequently operationalized. Domain analysis further shows concentration in healthcare, critical systems, and industrial applications, with limited real-world validation or longitudinal evaluation. The study identifies key gaps, including the absence of end-to-end RE4HCAI processes, weak formal modeling of human-centered requirements, limited treatment of data and model-centric requirements, and inadequate validation practices and outlines future research directions for building rigorous, value-aligned, and operationalizable Human-Centered RE for AI systems.

Index Terms—Human-Centered Artificial Intelligence, Human-Centered Design, Requirements Engineering, Human-Centered Design, Non-Functional Requirements, Explainability, Transparency, Trust, Participatory Design, RE4AI; Systematic Mapping Study

I. INTRODUCTION

Software projects continue to fail at alarming rates, often due to persistent Requirements Engineering (RE) challenges such as unclear goals, inadequate communication, and incomplete or misunderstood requirements, as evidenced by large-scale studies such as NaPiRE [1]. As artificial intelligence (AI) becomes increasingly embedded in critical decision-making domains, these longstanding RE issues take on amplified significance. Failures in AI systems have produced real-world harm, including racial disparities in healthcare risk prediction [2], intersectional accuracy gaps in facial analysis systems [3], and fairness and accountability concerns in recidivism prediction tools [4]. These cases demonstrate that AI failures are rarely technical mishaps alone they often arise from missing, ambiguous, or poorly articulated requirements related to ethics, fairness, transparency, trust, and human impact.

These failures motivate the growing emphasis on *Human-Centered Artificial Intelligence* (HCAI), an evolving paradigm

that seeks to ensure AI technologies augment rather than replace human capabilities while aligning with human values, societal norms, and ethical constraints. As framed by Shneiderman [5], HCAI can be understood both as a *goal* of the AI systems that are safe, reliable, and supportive of human well-being and as a *process* which is a set of human-centered practices that guide the design, development, and deployment of AI. Achieving HCAI thus requires more than functional accuracy or system performance; it requires processes grounded in Human-Centered Design (HCD), involving users in the process, capturing human values, evaluating social implications, and ensuring transparency and control throughout the AI lifecycle.

Requirements Engineering plays a foundational role in realizing this vision. Traditionally, RE focuses on eliciting, analyzing, specifying, and validating requirements to align system behavior with stakeholder needs. Although RE is often assumed to be inherently human-centered, prior research shows that conventional RE practices tend to emphasize technical and business concerns over experiential, ethical, and value-sensitive dimensions. With AI systems introducing new challenges such as data-driven uncertainty, opacity, emergent behavior, and value-laden impacts, there is a growing need to expand RE practices to explicitly incorporate fairness, explainability, trustworthiness, and human agency as first-class requirements.

Despite increasing attention to HCAI and an expanding body of work on Requirements Engineering for AI (RE4AI), there remains limited consolidated understanding of how human-centered RE practices are being applied to AI systems. Specifically, gaps persist in identifying (1) which RE stages are addressed, (2) what human-centered processes, methods, and artifacts are used, (3) which non-functional requirements (NFRs) tied to HCAI are captured, and (4) which domains adopt such practices. Existing reviews focus either on ethics in AI or RE4AI broadly but do not synthesize how RE contributes to achieving HCAI principles through user involvement, human values, and human-centered processes.

To address this gap, we conduct a Systematic Mapping Study (SMS) with the following research questions, (RQ1) What RE stages (elicitation, analysis, specification, validation) are involved in RE4HCAI studies, (RQ2) What processes, methods, techniques, frameworks, or artifacts are used or generated for HCAI in RE, (3) Which NFRs are addressed by RE4HCAI studies, and (4) What are the domain areas in

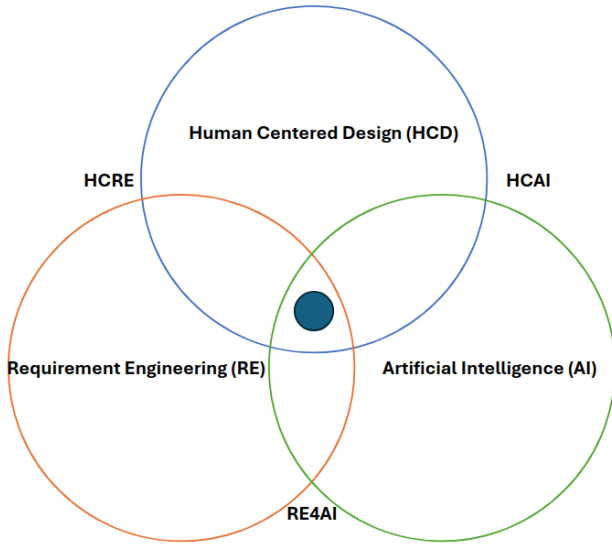


Fig. 1. A Visual Representation of the intersection of this study

which RE4HCAI is applied.

Several gaps were identified, including the dominance of early-stage elicitation activities, limited formalization and traceability of human-centered requirements, inconsistent definitions of key HCAI-related NFRs, and minimal attention to data, model, and MLOps-oriented requirements. The results further reveal weak validation practices and a concentration of work in only a few domains. These gaps motivate the need for more comprehensive, lifecycle-oriented, and empirically grounded RE approaches for AI. Accordingly, this SMS provides the basis for discussing these limitations and outlining future research directions aimed at developing rigorous, value-aligned, and operationalizable Human-Centered Requirements Engineering practices for AI systems.

II. BACKGROUND AND RELATED WORK

A. Human-Centered Design (HCD)

ISO [6] defines Human-Centered Design (HCD) as “an approach to interactive systems development that aims to make systems usable and useful by focusing on the users, their needs and requirements, and by applying human factors/ergonomics, and usability knowledge and techniques.” This definition highlights three essential principles: prioritizing user needs, enhancing usability, and grounding design decisions in human factors and ergonomics. ISO [6] further outlines six principles of HCD: (1) design is based on an explicit understanding of users, tasks, and environments, (2) users are involved throughout design and development, (3) design is driven and refined by user-centered evaluation, (4) the process is iterative, (5) the design addresses the whole user experience, and (6) the design team includes multidisciplinary skills and perspectives. The current study operates within the boundary defined by ISO’s HCD principles, and includes studies if they demonstrate at least two of these principles.

Earlier formulations of HCD emphasized similar ideas. Maguire [7] emphasized the active involvement of users throughout the design process and the careful balance between human and technological functions. Norman [8] similarly defined HCD as “an approach that puts human needs, capabilities, and behavior first, then designs to accommodate those needs, capabilities, and ways of behaving.” IDEO [9] underscores empathy as the cornerstone of effective design, where understanding users’ perspectives drives innovation. Across these definitions, a shared perspective emerges: users are not passive recipients, but active participants in shaping technology. HCD thus represents a philosophy where empathy, usability, and contextual understanding guide all design decisions.

Several processes operationalize HCD in practice. Participatory Design (PD) [10] is an approach rooted in democratic ideals, emphasizing the active involvement of users as equal partners in shaping the systems that affect their work and lives. It seeks to redistribute power in design processes, ensuring that users influence decisions rather than merely providing input. Co-design [11], in contrast, refers to the creative collaboration of designers and non-designers throughout the design process, using generative tools and shared ideation methods to imagine new possibilities. While PD is grounded in political empowerment, co-design focuses on collective creativity and innovation. Design Thinking [12] provides a process-oriented and innovation-driven framework, integrating empathy with iterative prototyping and balancing desirability (human value), feasibility (technical viability), and viability (business context).

HCD is also shaped by a range of related design traditions and theoretical foundations. User-Centered Design (UCD) [8] is a direct precursor to HCD, with strong emphasis on usability and cognitive principles such as mental models, affordances, and the human action cycle. Friedman’s Value Sensitive Design (VSD) [13] extends HCD by embedding ethical and moral considerations into the design process. Research in human factors and ergonomics [14] provides the scientific basis for HCD, focusing on optimizing interaction across physical, cognitive, social, and organizational dimensions.

B. Artificial Intelligence (AI)

The OECD (2019) [15] defines Artificial Intelligence (AI) as a machine-based system capable of influencing its environment by producing outputs, such as predictions, recommendations, or decisions based on a set of predefined objectives. AI systems use machine- and/or human-generated data to (1) perceive real or virtual environments, (2) abstract these perceptions into models through automated processes such as machine learning or through engineered rules, and (3) use model inference to formulate possible courses of action.

More broadly, AI refers to computational techniques designed to perform tasks that typically require human intelligence, including reasoning, pattern recognition, learning from experience, and adaptive decision-making [16]. Traditional approaches relied on symbolic rules and expert knowledge,

whereas contemporary systems increasingly use data-driven methods most notably machine learning and deep learning to identify patterns and optimize performance with minimal explicit programming. AI is thus best understood not as a single technology, but as a collection of methods that enable machines to interpret information, learn from it, and act autonomously or semi-autonomously in ways aligned with human-defined goals.

C. Human-Centered AI

The question of what constitutes *Human-Centered Artificial Intelligence (HCAI)* has become increasingly significant. Two notable works, [17] and [18], offer influential attempts to define and conceptualize HCAI. The study by [17] provides a formal definition and theoretical grounding for the concept, while [18] presents a comprehensive literature review that organizes existing HCAI research into four key dimensions. Table I is reproduced from [17] with our understanding of NFRs from the definition added to that table, compiles various definitions of HCAI proposed in the literature.

According to [17], HCAI focuses on understanding human purposes, values, and desired AI properties in the creation of AI systems by applying Human-Centered Design (HCD) principles. The goal of HCAI is to augment human capabilities while preserving human control over AI systems. This requires attention to the necessity and context of the system, as well as its ethical and legal implications, to support both individual and societal well-being. [17] organizes HCAI into three spheres: *purpose*, *values*, and *AI properties*. The *purpose* sphere emphasizes *augmentation* rather than *automation*. The *values* sphere refers to the fundamental beliefs that guide human judgment and behavior, classified into Ethical Values, Protection Values, and Performance Values. The third sphere, *AI properties*, includes attributes such as Accountability, Controllability, Responsibility, Sustainability, Explainability, Intellegibility, Transparency, Reliability, Robustness, Trustworthiness, and Traceability.

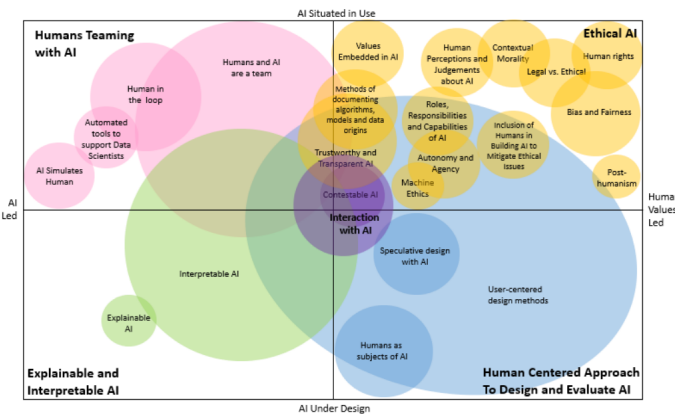


Fig. 2. Map of the field of HCAI picked from [18]

Similarly, [18] identifies four primary aspects of HCAI: Human Teaming with AI, Ethical AI, Explainable and Interpretable AI, and Human-Centered Approaches to Design

and Evaluate AI. The *Ethical AI* dimension encompasses issues such as contextual morality, values, trustworthy and transparent AI, autonomy and human agency, machine ethics, legal considerations, and fairness and bias. The *Human-Centered Approaches to Design AI* category highlights the use of Human-Computer Interaction (HCI) methods and tools to design and evaluate AI systems. These methods include a wide range of techniques for understanding human behavior and needs, organized into specific methods, broader domain methods, field study methods, and evaluations involving end users.

Ben Shneiderman [5] conceptualizes HCAI from two complementary perspectives: as a *process* or *goal*. In this review, emphasis is placed on understanding the processes that contribute to achieving the goals of HCAI.

To operationalize many of the ideas associated with HCAI, Non-Functional Requirements (NFRs) for machine learning systems offer a practical lens. A systematic compilation of NFRs [19] identifies 35 widely recognized attributes, including Accuracy, Adaptability, Bias, Completeness, Complexity, Consistency, Correctness, Domain Adaptation, Efficiency, Ethics, Explainability, Fairness, Fault Tolerance, Flexibility, Integrity, Interpretability, Interoperability, Justifiability, Maintainability, Performance, Portability, Privacy, Reliability, Repeatability, Retainability, Reproducibility, Reusability, Safety, Scalability, Security, Testability, Transparency, Traceability, Trust, and Usability. Similarly, [20] groups several of these attributes into broader categories such as Accuracy, Maintainability, Resiliency, Sustainability, and Usability.

Although many of these NFRs originate from traditional software engineering, the increased adoption of AI has amplified the importance of attributes related to ethics, trust, and transparency. Table II organizes NFRs into Human-Centered and Non-Human-Centered groups based on the classifications provided in [17] and [18]. Notably, some NFRs in the Non-Human-Centered category may, depending on context, be interpreted as Human-Centered.

Overall, HCAI can be understood as a subset of AI that emphasizes the integration of human values, needs, and oversight throughout the AI lifecycle. Unlike conventional AI approaches that often prioritize automation and optimization, HCAI focuses on augmenting human capabilities and ensuring ethical, transparent, and user-aligned system behavior

D. Requirement Engineering (RE)

Requirements Engineering (RE) is the disciplined process of identifying, analyzing, documenting, validating, and managing the requirements of a software system. According to SWE-BOK [21], RE comprises four core activities: elicitation, analysis, specification, and validation (also referred to as verification and validation). Elicitation involves acquiring requirements from stakeholders; analysis focuses on refining, negotiating, and prioritizing those requirements; specification produces formal or semi-formal representations such as Software Requirement Specifications (SRS); and validation ensures the

requirements accurately reflect stakeholder needs and system constraints.

Foundational texts such as Sommerville [22] stress that RE sits at the heart of the Software Development Life Cycle (SDLC), influencing subsequent phases including design, implementation, testing, and maintenance. While traditionally associated with plan-driven development, RE also plays a critical role in Agile methodologies, where requirements are captured through lightweight, iterative artifacts such as user stories, backlogs, and sprint reviews [23]. Agile reframes RE as a continuous, evolving process rather than a discrete early stage, emphasizing customer collaboration and rapid feedback loops.

Despite methodological differences across development models, RE consistently seeks to understand the goals, constraints, and rationale behind a system. This forms the foundation upon which human-centered perspectives can later be integrated.

E. Human Centered RE

Although RE is often assumed to be inherently human-centered given its focus on stakeholders, needs, and context traditional RE practices have historically emphasized technical, functional, and business concerns over human experiences. Early work in Human-Centered Requirements Engineering (HCRE) [24] demonstrated that classical RE approaches frequently overlooked user goals, concerns, emotions, and values. By analyzing tasks, agents, and user intentions, HCRE highlighted discrepancies between user needs and system-driven specifications.

RE research increasingly recognizes the importance of integrating human factors across all RE activities. Grundy et al. [25] conducted a systematic review revealing that RE is influenced by human factors at individual (cognitive and behavioral), team (communication and collaboration), and organizational (culture and structure) levels. A subsequent study [26] developed a prioritization framework demonstrating how practitioners value human aspects in RE decision-making. However, it is important to distinguish human factors in RE from Human-Centered Design (HCD). While human factors emphasize developer coordination and teamwork, HCD prioritizes end-user needs, usability, desirability, and stakeholder values as primary outcomes.

Human-centered RE extends beyond isolated user engagement. It requires continuous stakeholder integration, explicit modeling of human values and contexts, and participatory and iterative practices. As Ahmad et al. [27] note, RE involves understanding “the need, desire, or want to be satisfied” as well as identifying stakeholders and the rationale for their involvement. Truly human-centered RE incorporates qualities such as trust, fairness, explainability, transparency, and usability as first-class requirements rather than afterthoughts.

HCD-inspired methods strengthen RE by providing systematic ways to understand people and contexts. ISO [28] distinguishes between methods (structured means for performing activities), techniques (systematic procedures), and artifacts

(tangible outputs). Prior research [29] and design frameworks such as the LUMA System [30] highlight elicitation and modeling techniques including interviews, surveys, workshops, scenarios, prototyping, journey mapping, personas, and mind mapping that improve empathy, shared understanding, and early discovery of user needs. Artifacts such as personas, journey maps, scenarios, storyboards, wireframes, and SRS documents bridge communication between users and designers by making implicit expectations explicit.

F. RE4AI

Requirements Engineering for Artificial Intelligence (RE4AI) is not a new domain; instead, it represents the consolidation of long-standing challenges associated with specifying, analyzing, and validating AI-intensive systems. Two widely cited review papers, [31] and [27], provide foundational mappings of this research landscape. [31], using SWEBOK [21] as the reference model for RE processes, mapped 126 studies and found that most RE4AI research concentrates on elicitation and analysis. Their review identifies numerous models, techniques, and processes proposed for AI-related RE, along with recurrent challenges such as difficulty in specifying requirements, inconsistent definitions of explainability, the rise of data as a new type of requirement, limited human-centered approaches, gaps between ML engineers and end users, uncertainty inherent in ML models, classification challenges, and legal and ethical constraints.

[27] similarly highlights frameworks and modeling approaches for RE4AI, identifying issues such as reliance on assumptions, overconfidence in AI systems, emerging data requirements, and gaps between various stakeholders. This aligns with earlier work such as [32], which emphasized that AI systems require data-aware requirements addressing concerns such as explainability, fairness, discrimination, and legal compliance.

More recent work [33] integrates industrial guidelines with academic literature to understand how practitioners document AI requirements. Their findings reveal five human-centered needs for AI systems: user needs, model needs, data needs, feedback and control, explainability and trust, and error and failure handling.

Overall, RE4AI requires expanding traditional requirements thinking to incorporate data considerations, ML model behavior, uncertainty management, explainability, human values, and ethical constraints. While conventional RE focuses on defining stable functional and non-functional requirements, RE4AI must address dynamic, probabilistic, and value-laden aspects of AI systems. The incorporation of Human-Centered RE principles is therefore critical, ensuring that AI systems not only meet technical objectives but also remain aligned with user expectations, societal values, and responsible innovation practices.

TABLE I
DEFINITIONS OF HUMAN-CENTERED ARTIFICIAL INTELLIGENCE (HCAI) AND ASSOCIATED NON-FUNCTIONAL REQUIREMENTS

Definition	NFRs Touched Upon in the definition	Citation
HCAI focuses on amplifying, augmenting, and enhancing human performance in ways that make systems reliable, safe, and trustworthy. These systems also support human self-efficacy, encourage creativity, clarify responsibility, and facilitate social participation.	Reliability, Safety, Trustworthiness	[34]
HCAI includes three main components: (1) ethically aligned design (fairness, justice, non-replacement of humans), (2) technology reflecting human intelligence, and (3) human factors design ensuring explainability, comprehensibility, usefulness, and usability.	Fairness, Ethics, Explainability, Usability	[35]
HCAI aims to augment human abilities, address societal needs, and draw inspiration from human beings.		[36]
HCAI is a perspective that intelligent systems must be designed with awareness that they are part of a larger system consisting of human stakeholders such as users, operators, and clients.		[37]
HCAI design requires three integrated perspectives: rationalistic (technology), humanistic (people), and judicial (policies).		[38]
Human-centered means that a system should always have the human partner as part of its deliberation, performing tasks for someone, in some context, and being aware of its social and ethical consequences.	Ethics	[39]
AI systems need to be human-centric, committed to the service of humanity and the common good, aiming to improve human welfare and freedom.	Ethics	[40]
HCAI is a synergistic approach aligning AI with human values, ethical principles, and legal requirements to ensure safety, security, and trustworthiness.	Ethics, Legality, Safety, Security, Trustworthiness	[41]
HCAI involves designing AI systems that enhance human capacities and experiences rather than replacing them through automation.		[42]

TABLE II
HUMAN-CENTERED VS NON-HUMAN CENTERED NFRs FOR AI/ML SYSTEMS

Non-Human-Centered NFRs	Human-Centered NFRs
Accuracy, Availability, Adaptability, Capacity, Completeness, Complexity, Consistency, Correctness, Domain Adaptation, Efficiency, Fault Tolerance, Flexibility, Integrity, Interoperability, Maintainability, Performance, Portability, Repeatability, Retainability, Reproducibility, Reusability, Replaceability, Stability, Scalability, Transferability, Testability, Traceability	Accountability, Bias, Controllability, Responsibility, Sustainability, Ethics, Explainability, Fairness, Intelligibility, Justifiability, Privacy, Transparency, Reliability, Robustness, Trustworthiness and Traceability, Usability, Security, Safety, Predictability

III. SYSTEMATIC MAPPING STUDY

We are conducting an Systematic Mapping Study to understand how Human Centered RE is being used in AI. We are looking at processes that involve the end user to achieve HCAI goals. Although there is growing work on HCAI, there is limited synthesis of how RE contributes to the principles of HCAI. We have three major goals while conducting this Mapping study. These are to identify RE processes, methods, techniques, framework or artifacts proposed or used. The NFRs that are addressed and the application domain to which it is applied.

A. RQs

RQs for the mapping study.

- **RQ1:** What RE stages (elicitation, analysis, specification, validation) are involved in RE4HCAI studies?
- **RQ2:** What processes, methods, techniques, frameworks, or artifacts are used or generated for HCAI in RE?
- **RQ3:** Which NFRs are addressed by RE4HCAI studies?

- **RQ4:** What are the domain areas (e.g., healthcare, education, autonomous systems) in which RE4HCAI is applied?

TABLE III
PICO-C DEFINITION FOR THE SYSTEMATIC MAPPING STUDY

Component	Description
Population (P)	Research studies involving Requirements Engineering activities for Artificial Intelligence systems that incorporate human-centered principles.
Intervention (I)	RE methods, techniques, processes, frameworks, or artifacts supporting Human-Centered AI development.
Comparison (C)	Not applicable. No direct comparison between interventions is performed; studies may be grouped/classified across RE stages, NFRs, and domains.
Outcome (O)	Identification and classification of: (1) RE stages addressed, (2) RE methods/techniques/frameworks/artifacts used or produced, (3) NFRs addressed, (4) domains of application.
Context (C)	Peer-reviewed English-language research published from 2000 onward, focusing on intelligent, autonomous, AI, or ML-based systems, retrieved mainly from ACM DL and IEEE Xplore.

B. Search Strategy

We conducted a search on the following digital libraries: **IEEE Xplore** and **ACM Digital Library**. The search string is an intersection of RE, HCD and AI. The search string has three parts which are:

- **C1:** ("requirements engineering" OR "requirements elicitation" OR "requirements analysis" OR "requirements specification" OR "requirements validation" OR "requirements management" OR "agile")
- **C2:** ("human-centered" OR "user-centered" OR "human-centric" OR "participatory design" OR "co-design" OR

"value-sensitive design" OR "human factors" OR "user experience" OR "design thinking")

- **C3:** ("artificial intelligence" OR "AI" OR "machine learning" OR "intelligent system*" OR "autonomous system*" OR "ML" OR "deep learning")

The search string is an AND operation of all of the above three and the search will be conducted on the abstract of the paper. For IEEE Xplore we use 'all metadata'. The search is conducted on papers that are written post 2000. We limit our search to just journal or conference papers. For ACM we are searching based on title, abstract and keyword. This was followed by snowballing of papers.

C. Inclusion and Exclusion Criteria

1) Inclusion Criteria:

- **IC1:** Studies addressing at least one stage of Requirements Engineering.
- **IC2:** Studies incorporating Human-Centered, User-Centered, or Human-Factors principles.
- **IC3:** Studies focusing on AI, ML, or intelligent/autonomous systems.
- **IC4:** Studies that propose, adapt, or evaluate a process, technique, framework, method, or artifact.
- **IC5:** Peer-reviewed and written in English.

2) Exclusion Criteria:

- **EC1:** Papers not related to RE for HCAI.
- **EC2:** Studies focusing only on functional requirements.
- **EC3:** Editorials, posters, tutorials, literature reviews, or non-scholarly grey literature.
- **EC4:** Duplicates or earlier versions of included papers.
- **EC5:** Studies unrelated to AI contexts (e.g., general HCI or UX-only studies).
- **EC6:** Not written in English.

D. Data Extraction

The process of data extraction followed the protocol proposed by Kitchenham et al. [44]. For each paper retrieved based on the defined search criteria, a set of attributes was captured, including type of paper, UniqueID, Author(s), Editor(s), Title of the article, Pages, Keywords, DOI, Year of publication, ISBN, Publisher, Extraction date, and Database. Our search on ACM gave 55 papers out of which 7 papers were selected for the final study. IEEE Xplore gave 218 papers out of which 20 papers were selected. 4 papers were snowballed. In total 31 papers were reviewed for this study.

IV. RESULTS

A. RQ1: What RE stages (elicitation, analysis, specification, validation) are involved in RE4HCAI studies?

Out of the 31 studies analyzed, the RE stages exhibit a clear emphasis on early-phase human-centered activities, with elicitation emerging as the dominant focus. Specifically, 28 studies (90%) incorporate elicitation, underscoring its role as the primary entry point for embedding user values, ethical considerations, and contextual needs in AI requirements. This

TABLE IV
PAPER SELECTION SUMMARY BY DIGITAL LIBRARY

Category	IEEE Xplore	ACM DL
Total	218	55
Included	20	7
Not Accessible	0	8
EC1	179	27
EC2	0	0
EC3	9	12
EC4	2	0
EC5	8	1
EC6	0	0

is followed by analysis (16 studies, 52%), which often refines elicited inputs through modeling and evaluation; validation (12 studies, 39%), emphasizing iterative feedback and testing; and specification (8 studies, 26%), the least addressed stage, typically involving formal artifact generation for traceability. 4 studies involve all 4 phases together as they a part of the agile process.

There are certain aspects due to which these stages have been classified accordingly. Elicitation involves actively engaging users to surface their values, goals, concerns, and other human-centered non-functional requirements (NFRs). Analysis then interprets these elicited values into concrete constraints, goals, and trade-offs, while also identifying mismatches between user mental models and system behavior, as well as potential risks, harms, or ethical tensions. During specification, these insights are formally encoded as explicit human-centered NFRs, accompanied by clear documentation of the reasoning behind each requirement. Finally, validation ensures that the system and its human-centered NFRs are evaluated with real users to confirm that the captured requirements accurately reflect their needs and expectations.

B. RQ2: What processes, methods, techniques, frameworks, or artifacts are used or generated for HCAI in RE?

In the 31 studies, we analyzed a diverse array of processes, methods, techniques, frameworks, and artifacts that emerge to support RE4HCAI. These elements prioritize user involvement, iterative refinement, and the integration of ethical, contextual, and value-driven considerations, aligning with HCD principles such as empathy, usability, and multidisciplinary collaboration.

The processes in these RE4HCAI studies are predominantly iterative and collaborative, drawing on HCD traditions to embed human values early and sustain them throughout the AI lifecycle. Participatory Design (PD) and Co-Design dominate, appearing in 7 studies, as they facilitate democratic stakeholder engagement and collective creativity core to HCAI's emphasis on human control and well-being. For instance, PD is employed for ethical card-based prototyping [47] and full-lifecycle integration [48], [71], often yielding insights into trust and transparency in critical domains. Co-Design, extends this by focusing on shared workshops for diverse user groups, such as parents in child protection AI [50] or elderly in ambient assistive tech [51], [52]. Agile processes are ubiquitous (6

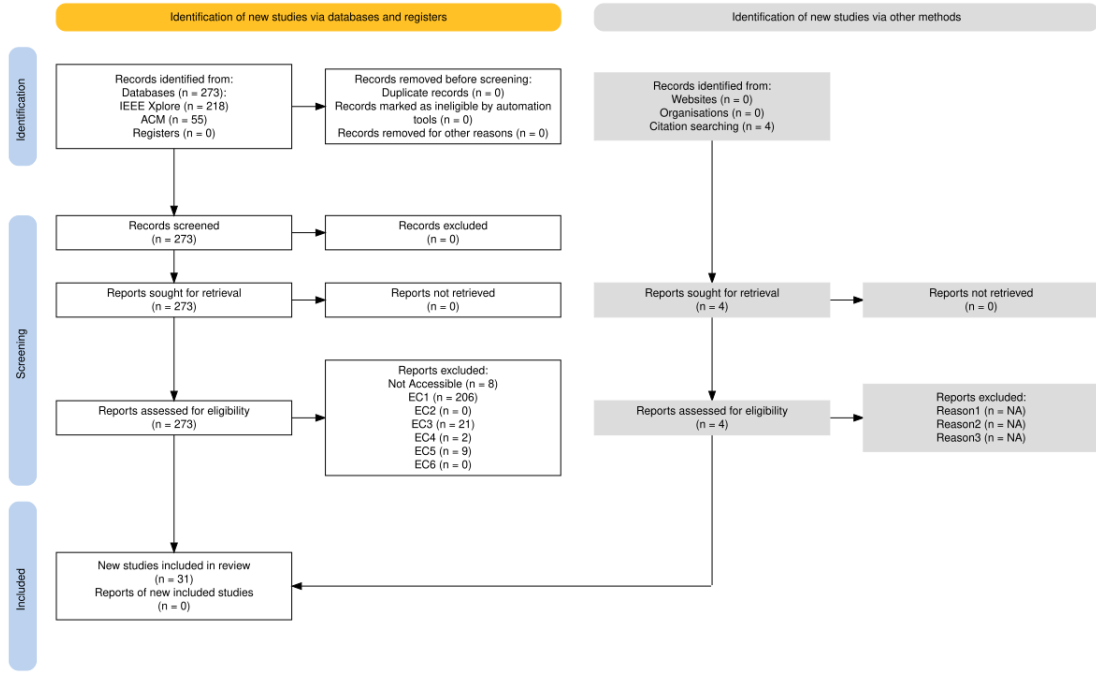


Fig. 3. PRISMA diagram of the study generated by [43]

TABLE V
RE STAGES WITH COUNT

RE Stage	Count	Citation	How It Is Conducted
Elicitation	28	[45]–[72]	Card Based Methods, Scenario Based Methods, gamified and participatory workshops, co-design sessions, living-lab workshops, stakeholder identification, interviews, observations, AEIOU analysis, value modeling, user modeling, HMW sessions, prototyping sessions
Analysis	16	[49], [50], [52], [55], [56], [58], [60], [62], [63], [65]–[68], [73]–[75]	Scenario detailing and stakeholder review, DRSM reviews, use case and goal modeling, multi-aspectual analysis, ethical-pedagogical and interactional analysis, motivational goal modeling, LLM-based agent critique loops, cognitive and mental-model analysis, thematic analysis, risk analysis and scenario modeling
Specification	8	[50], [52], [56], [60]–[64]	User Story, co-design based specification, motivational goal models, modular microservice specifications, use-case diagrams, storyboards, wireframes, hierarchical task analysis and trust-state charts
Validation	12	[49]–[51], [53]–[55], [59]–[63], [74]	Stakeholder review, iterative co-design evaluations, living-lab testing, simulation-based validation, rated user feedback, user studies and feedback loops, prototype evaluation and UI walkthroughs, scenario-based performance evaluations, usability testing

studies), promoting sprint-based user feedback and adaptability; examples include ethical user story generation [45] and hypertension management prototypes [54]. Design Thinking recurs in 7 studies, structuring empathy-driven ideation for domains like risk communication [62] and IoT [70]. Integrated approaches, hybridize these for robustness e.g., Agile + Co-Design + Design Thinking in living labs [51] or Agile + Participatory Design in smart workplaces [71], [72]. These hybrids address HCAI gaps like stakeholder silos, ensuring augmentation over automation.

Methods and techniques operationalize processes at a granular level, capturing human purposes and values. Workshops

(e.g., Co-Design, gamified, scenario-based) are the most common, enabling contextual exploration in domains like health-care [46], [54] and finance [49]. These often incorporate techniques like empathy mapping or role-playing to uncover ethical nuances. Card-based and scenario-based techniques elicit ethical/explainable needs via narratives [46], [47], using prompts for value conflicts and simulations of controllability gaps. Interviews and surveys ground value modeling [58], [66], combined with thematic analysis to cluster NFRs like privacy. Goal-oriented methods extend RE for motivational alignment [56], [73], integrating theories like Schwartz's val-

TABLE VI
PROCESSES USED ACROSS STUDIES

Process	Studies Used in	How It Was Used
Participatory Design (PD)	[47], [48], [71], [72]	Ethical card elicitation, gamified prototyping, full-stage empathy integration
Co-Design	[50]–[52]	Collaborative workshops for user journeys, scenario co-creation, stakeholder empathy
Agile	[45], [50], [54], [60], [63], [72]	Sprint-based user stories, iterative feedback loops, ethical requirement refinement
Design Thinking	[51], [54], [61]–[63], [70], [71]	Empathy mapping, HMW questioning, journey ideation for contextual needs

ues to map emotional goals against trustworthiness. Cognitive and mental modeling addresses explainability [55], [65], using ontologies to capture user mental models. Other techniques include Hierarchical Task Analysis for trust decomposition [64], AEIOU frameworks for observational elicitation in safety-critical settings [62], and affinity diagramming for synthesis [70]. Prototyping and simulations test usability iteratively [53], [60], [61], with gamified probes for engagement [48]. Artifacts, make implicit needs explicit for HCAI traceability. User stories lead often ethical or scenario-infused [45], [52], [61], framing requirements as value-aligned narratives. Scenarios and use cases, visualize risks [46], [49], [62]. Personas and prototypes foster empathy [60], [67], [72], enabling feedback loops. Other outputs include motivational goals [56], affinity diagrams [70], and state charts [64], emerging mainly from analysis/specification to formalize.

Explicit frameworks appear in 5 studies, providing structured models for HCAI properties like ethics and explainability. The FEPER framework [74] integrates pedagogical/ethical alignment for educational AI, emphasizing value-sensitive interactions through efficiency metrics and feedback loops. AI-EIA [59] extracts guideline-based ethical requirements, supporting societal well-being assessments via impact scoring techniques. AENEA [61] co-creates stakeholder needs for sustainable urban AI via design thinking workshops and prototyping. DoReMi [69] embeds value-sensitive design in HCD for clinical explainability, with components for rationalistic, humanistic, and judicial perspectives. [68] proposes a modeling framework for human-centered elicitation in VR apps, using interview-driven tools for transparency and trust specification.

C. RQ3: Which NFRs are addressed by RE4HCAI studies?

To delineate the human-centered orientation of non-functional requirements (NFRs) in RE4HCAI, we first classified them into human-centered (e.g., usability, trust, explainability) versus non-human-centered (e.g., efficiency, adaptability) categories, as outlined in Table II. This framework guided our analysis of the 31 studies. Usability dominates as the most recurrent NFR (15 studies, 48%), a continuity from

TABLE VII
PROPOSED OR USED FRAMEWORKS IN RE4HCAI STUDIES

Framework Alias	Citation	Core Components/Usage
FEPER	[74]	Human-AI efficiency, ethics, pedagogy via interaction modeling
AI-EIA	[59]	Ethical extraction from guidelines; impact assessment
AENEA	[61]	Stakeholder co-creation workshops for sustainability
DoReMi	[69]	HCD components with value-sensitive design for explanations
HC-RE Framework	[68]	Elicitation/modeling tool for human-centered AI reqs

traditional RE. In HCAI contexts, it manifests through participatory prototyping and journey mapping to ensure intuitive interactions, as in ambient assistive technologies [51], [54]. Emerging AI-specific concerns trail closely: transparency (11 studies, 35%), vital for demystifying opaque models via goal-oriented elicitation [47], [56], [73]; explainability (9 studies, 29%), operationalized in scenario-driven designs for clinical interpretability [46], [48], [69]; and trust (8 studies, 26%), cultivated through simulations and value-sensitive modeling to avert overreliance [53], [64], [74]. Ethics (6 studies, 19%) and privacy (5 studies, 16%) further underscore the moral imperatives of HCAI. NFRs, such as accessibility (4 studies, 13%), fairness (4 studies, 13%), and flexibility (2 studies, 6%), appear sporadically, often implied rather than explicitly formalized, while outliers like reliability and safety (1 study each) surface in niche critical applications [53], [62]. Table VIII enumerates these NFRs with their frequencies and exemplars, affirming that over 70% of studies engage multiple human-centered qualities, predominantly in elicitation and analysis phases.

TABLE VIII
NFRs

NFR	Count	Citation
Ethics	6	[45], [47], [58], [59], [73], [74]
Usability	15	[45], [49]–[51], [54], [55], [60], [62], [63], [67], [68], [70]–[72], [74]
Explainability	9	[46], [48], [57], [60], [65], [68], [69], [74], [75]
Transparency	11	[47], [48], [56], [58], [61], [63]–[66], [68], [73]
Trust	8	[48], [53], [54], [56], [64], [66], [68], [74]
Accessibility	4	[51], [60], [62], [67]
Fairness	4	[52], [56], [58], [73]
Equality	1	[52]
Reliability	1	[53]
Privacy	5	[54]–[56], [58], [66]
Adaptability	1	[55]
Efficiency	1	[55]
Accountability	1	[58]
Safety	1	[62]
Flexibility	2	[63], [72]



Fig. 4. Word Cloud Representation of the NFRs used

D. RQ4: What are the domain areas (e.g., healthcare, education, autonomous systems) in which RE4HCAI is applied

RE4HCAI manifests across a spectrum of domains, underscoring its adaptability to contexts where ethical, usability, and trustworthiness imperatives are acute. Among the 31 studies, healthcare stands out as the most prevalent application area (6 studies, 19%), with a particular emphasis on vulnerable populations. For elderly users, RE4HCAI prioritizes monitoring seniors' activities of daily living [46] and ambient assistive living technologies [51], integrating participatory methods to elicit privacy-sensitive requirements. Similarly, pediatric contexts receive attention, as seen in security-focused educational apps for children [50] and clinical decision support systems tailored to child health [69], where co-design ensures age-appropriate explainability and safety. Critical systems represent another focal point (3 studies, 10%), addressing high-stakes environments through scenario-based and simulation-driven approaches. Applications include aircraft condition-based maintenance [48], UAV autonomy in combat scenarios [53], and risk communication in mining operations [62], all leveraging gamified elicitation to embed resilience and trust in AI requirements. Finance (1 study, 3%) and human resources (1 study, 3%) emerge as targeted yet underrepresented domains, with efforts centered on fairness in credit detection [49] and inclusive hiring platforms [52] to counteract algorithmic biases. Complementing these domain-specific explorations, 11 studies (35%) develop general frameworks and tools, such as ethical card-based methods [45], [47] and value-annotated goal models [56], offering scalable foundations for broader AI adoption. Notably, a recurring pattern surfaces: RE4HCAI disproportionately targets critical scenarios and at-risk demographics, elders, children, and end-users in life-impacting systems, over routine commercial applications.

V. DISCUSSION AND GAPS

This mapping study consolidates how Human-Centered Requirements Engineering (HCRE) is currently applied within AI system development. The findings point to meaningful progress toward embedding human values in AI requirements, yet they also reveal several gaps that limit the maturity, consistency, and practical impact of RE4HCAI.

TABLE IX
DOMAIN

Domain	Count	Description
General	11	[45], [47], [55]–[57], [59], [64]–[66], [73], [75]
Healthcare	6	Tracking Seniors' activities of daily living [46], Ambient Assistive Living [51], Hypertension management and Remote Patient Management [54], patient monitoring, personalized diet and health recommendations based on biomarkers [60], Therapy and Diagnosis [67], children healthcare [69]
Critical Systems	3	Aircraft Maintenance [48], UAVs in Combat Scenario [53], Mining Operations [62]
Finance	1	Credit Detection System [49],
Education	2	Security-knowledge app for parents and kids [50], Chatbots and Social Robots in Educational Setting [74]
Industrial	3	Industry 4.0 [70], IT Solution Architecture [71], Data Science and Process Automation [72]
Others	5	HR Hiring Systems [52], Fake Review Detection [58], Ecological Aquatic Urban Systems [61], Personalized product recommendation UX [63], VR Deep Learning App [68]



Fig. 5. Word Cloud Representation of the domains addressed

A. Emphasis on Early-Stage RE Activities

The majority of studies focus on early RE activities, with 28 papers addressing elicitation and 16 addressing analysis. Workshops, co-design sessions, scenario-based elicitation, and value-oriented interviews dominate this phase. However, the systematic transition of these insights into later RE stages is limited. Only a small subset of studies engage in formal specification (8 papers) or structured validation (12 papers). As a result, human-centered insights though rich are seldom formalized, operationalized, or validated, reducing their traceability across the AI lifecycle and diminishing their influence on deployed systems. Three gaps can be identified here. One that there is a lack of end-to-end RE4HCAI process. There is a high concentration on elicitation, while few studies provide structured pathways from values to formal requirements, design decisions, and implementation constraints. The second gap is that there is weak formalization and traceability. Most studies produce qualitative insights but few generate structured artifacts that carry human-centered requirements forward. The

third is that their is limited validation of human-centered requirements for AI. Most of the validation techniques are superficial and there is no guarantee that the outputs expected at validation with scenarios and prototyping techniques mentioned would be the actual output from the AI black box when implemented.

B. Fragmented Treatment of Human-Centered NFRs

Across the reviewed work, usability, transparency, and explainability emerge as the most commonly addressed NFRs. In contrast, key HCAI-related qualities including fairness, accountability, robustness, and safety appear sporadically or implicitly. A deeper issue lies in the absence of standardized definitions or ontologies for these NFRs. For instance, "trust," "explainability," and "transparency" are interpreted differently across studies, complicating comparison and weakening requirements consistency. Moreover, few studies propose measurable indicators or metrics for evaluating these NFRs, leaving them open to subjective interpretation and limiting verification potential during system evaluation. Two gaps identified here were that there is a lack of standardized definition across and that there is a lack of metric for evaluation.

C. Agile vs. Process-Driven Tensions

A notable pattern is the dichotomy between Agile-driven studies and process-heavy HCD frameworks. Agile promotes iterative, flexible elicitation but rarely supports stable documentation or formal modeling of NFRs such as fairness or accountability. Conversely, process-driven frameworks emphasize rigor and structure but are less commonly applied in real-world AI settings. The absence of integration strategies exacerbates the challenge of operationalizing HCAI, as teams struggle to reconcile iterative development with the rigor needed for ethically sensitive AI.

D. Domain Concentration and Limited Real-World Validation

Most reviewed studies cluster around healthcare, critical systems, and industrial domains areas that naturally emphasize safety, trust, and usability. However, there is limited evidence from public-sector AI, financial systems, e-commerce platforms, or high-scale deployed AI systems, despite their pressing societal impact. Additionally, validation is often lightweight, relying on early feedback or scenario reviews rather than longitudinal studies or deployed-system evaluations. As a result, it remains unclear whether the proposed human-centered requirements actually improve fairness, trust, or explainability in real-world systems.

E. Insufficient Integration of Data and Model Requirements

AI failures often stem from data issues (e.g., dataset bias, drift, representational harm) or model behavior (e.g., uncertainty, instability, opacity). Yet, only a minority of studies capture data-centric or model-centric requirements. While some papers introduce value modeling or multi-aspectual analysis, almost none integrate MLOps considerations such as dataset versioning, data drift detection, monitoring, or post-deployment retraining. This absence limits the completeness

of RE4HCAI and overlooks crucial lifecycle considerations. This is another gap that needs to be looked at.

VI. FUTURE WORK

Addressing the gaps above suggests several promising directions for future research:

A. End-to-End Human-Centered RE Processes

Future work should develop RE4HCAI processes that explicitly integrate elicitation, analysis, specification, validation, and maintenance. Such processes must support both Agile practices and structured RE, enabling value-driven requirements to persist throughout iterative development.

B. Formal Modeling of Human-Centered Requirements

There is a clear need for modeling languages, ontologies, and templates that capture human values, goals, emotions, ethical concerns, and HCAI-related NFRs (e.g., explainability, trust, fairness) at once. These models should support traceability and be compatible with existing RE tools.

C. Standardization and Metrics for HCAI NFRs

Future research should work toward unified definitions for HCAI-relevant NFRs and develop quantifiable metrics for properties such as fairness, transparency, accountability, and trustworthiness. This would significantly improve comparability, verifiability, and regulatory compliance.

D. Data, Model, and MLOps-Focused Requirements

Future work must extend RE4HCAI to encompass data-centric and model-centric requirements, including dataset representativeness, drift detection, monitoring pipelines, explainability constraints, and retraining strategies. Integrating MLOps practices into RE is crucial for long-term system reliability and safety.

E. Strengthening Validation of Human-Centered Requirements

Research should develop empirical evaluation frameworks, simulation environments, and longitudinal protocols to assess whether human-centered requirements lead to measurable improvements in user trust, fairness, decision quality, or reduction of harm in real-world deployments.

F. Human-Centered RE for Traditional Software Systems

A systematic review of human-centered RE practices in traditional software engineering may reveal transferable insights, helping bridge the conceptual gap between classical HCD practices and modern AI-driven RE.

VII. LIMITATIONS & THREATS TO VALIDITY

A. Search Bias

This study uses only IEEE Xplore and ACM DL, which may exclude relevant work published in other venues such as Springer, Elsevier, or arXiv. The search string may also miss papers that discuss RE or HCD implicitly without using the exact terminology. As a result, the final corpus may not fully represent the broader interdisciplinary space of RE4HCAI.

B. Extraction Bias

Data extraction required interpreting how each study mapped to RE stages, NFRs, processes, and artifacts. Because many papers do not explicitly use SWEBOK terminology or standardized HCD language, classification decisions relied on subjective judgment. This introduces the risk of inconsistent interpretation across studies.

C. Construct Validity

Key HCAI-related NFRs such as trust, explainability, transparency, and fairness lack standardized definitions and measurable criteria in the literature. Differences in how authors conceptualize these terms may affect how they were categorized in this study, limiting comparability and precision in the mapping.

D. Single Reviewer Bias

Screening, selection, and data extraction were conducted by a single reviewer, increasing susceptibility to subjective interpretation, confirmation bias, and potential oversight. Multi-reviewer processes with inter-rater agreement could reduce this risk in future work.

E. Snowballing Limitations

Snowballing depends heavily on the initially selected papers and may propagate their inherent biases. Relevant studies may still be overlooked if they do not cite or are not cited by the chosen seed papers. Thus, snowballing supplements but cannot fully compensate for database limitations.

VIII. CONCLUSION

This Systematic Mapping Study provides the first consolidated view of how Human-Centered Requirements Engineering (HCRE) is being applied in the development of AI systems. Across 31 studies, the findings highlight meaningful progress in integrating user needs, contextual understanding, and human values into early stages of AI system development. However, the review also exposes significant limitations that currently hinder the maturity and practical impact of RE4HCAI. Most research concentrates on elicitation and early analysis, with few studies offering structured pathways from human values to formal requirements, design decisions, or implementation constraints. As a result, human-centered insights often remain qualitative, weakly formalized, and insufficiently traceable across the AI lifecycle. Furthermore, critical NFRs central to HCAI such as fairness, accountability, robustness, and safety are inconsistently defined and rarely accompanied by measurable criteria. Domain coverage remains uneven, with limited evidence from large-scale, real-world AI deployments, and validation practices are typically lightweight or short-term. Finally, current RE4HCAI work pays insufficient attention to data-centric and model-centric requirements, despite their known role in AI failures.

Overall, the results underscore the need for end-to-end, lifecycle-oriented RE4HCAI approaches that integrate human-centered methods with formal modeling, standardized NFR

definitions, traceability mechanisms, and robust evaluation strategies. Advancing this area will require interdisciplinary collaboration, empirical grounding, and methodological innovation to ensure that AI systems are not only technically effective but also ethical, transparent, trustworthy, and aligned with human well-being.

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